

ScamShield

A presentation-ready safety tool for spotting scam and phishing messages before they cause harm.



Purpose

ScamShield helps users slow down and make a safer decision when an email, SMS, WhatsApp message, or other text feels suspicious.

Built for a reliable seven-minute GenAI classroom demonstration.

The numeric score is always produced by transparent local rules. Optional generative AI improves explanation wording only.

01 / AT A GLANCE

A calm answer to a high-pressure problem.

ScamShield turns a suspicious message into three things a user can understand quickly: a risk score, the evidence behind that score, and a safe next action.

MESSAGE INTAKE

Paste a message or choose one of five fictional demonstration samples.

EVIDENCE

See pressure tactics, credential requests, financial asks, impersonation, and URL signals.

NEXT ACTION

Get a clear recommendation based on LOW, MEDIUM, or HIGH risk.

What makes the demo trustworthy

- **Deterministic:** the same message produces the same score every time.
- **Local-first:** it works without an OpenAI API key or a security API.
- **Safe by default:** URLs are treated as untrusted text and never visited.
- **Traceable:** every result points back to visible indicators.

Demo checkpoint

The fictional Bank Account Warning sample returns **95% HIGH RISK**. The Safe Message sample returns **0% LOW RISK**.

02 / ARCHITECTURE

A simple pipeline with clear responsibilities.

The app is intentionally understandable for beginner and intermediate Python developers. Each stage does one job and passes evidence forward.



Project structure

File	Responsibility
app.py	Five-page Streamlit interface, navigation, samples, and presentation UI.
risk_engine.py	Weighted signals, score calculation, and LOW / MEDIUM / HIGH classification.
message_analyzer.py	Orchestrates language and URL checks and assigns a scam category.
url_analyzer.py	Parses visible URLs without opening or requesting them.
database.py	Creates SQLite storage, saves analyses, and seeds fictional history.
ai_explainer.py	Optional OpenAI wording with a deterministic local fallback.

03 / RISK MODEL

Evidence first. Explanation second.

ScamShield uses weighted pattern matching to create a 0–100 score. The score is never delegated to a language model, which keeps the result repeatable and auditable.

Risk level	Range	Meaning
LOW RISK	0–29	No major indicators detected. Continue to verify unexpected requests.
MEDIUM RISK	30–69	Suspicious patterns are present. Verify independently before taking action.
HIGH RISK	70–100	Multiple high-confidence signals are present. Do not respond or click links.

Signals the engine checks

Pressure	Credentials	Money	Identity
Urgent language Threats and consequences	Passwords Verification codes Bank or card details	Payments Gift cards Cryptocurrency Prize fees	Bank impersonation Government impersonation Package delivery impersonation

Important safety boundary

The app treats every URL as untrusted text. It may flag a URL based on visible characteristics such as an action-oriented path, a shortened host, or a non-HTTPS scheme, but it never connects to the destination.

04 / LIVE DEMO

A seven-minute walkthrough.

Use the following sequence to show the full product story without relying on an external API or live malicious content.

Time	What to show	Talking point
0:00–1:00	Dashboard	Point out seeded history, the risk distribution, and the four KPIs.
1:00–2:00	Analyze Message	Choose Bank Account Warning to load a fictional suspicious message.
2:00–3:00	Progress sequence	Show the five-stage investigation: read, language, URLs, score, recommendation.
3:00–4:15	AI Risk Analysis	Show the 95% HIGH RISK score, evidence cards, and highlighted phrases.
4:15–5:15	Investigation	Walk through Message Analyzer, URL Inspector, Risk Engine, and Safety Advisor.
5:15–6:15	Recommended Action	Emphasize not responding, not clicking, and verifying independently.
6:15–7:00	Safe Message	Run the safe sample to demonstrate the LOW result and the system's restraint.

Recommended sample sequence

01 / BANK WARNING

Expected result: 95% HIGH RISK

02 / SAFE MESSAGE

Expected result: 0% LOW RISK

03 / OPTIONAL

Try package, IRS, or prize examples to show MEDIUM results.

05 / RUN & OPERATE

Ready for a classroom demo.

ScamShield is designed to run locally in the Replit environment with no required API key and no external cybersecurity service.

Launch command

```
streamlit run scamshield/app.py --server.port 5000
```

Storage and optional AI

- SQLite history is created automatically at **scamshield/data/scamshield.db**.
- If the database is empty, the app seeds approximately 15 fictional analyses.
- Without an OpenAI key, the local rule-based Safety Advisor is used.
- With **OPENAI_API_KEY**, OpenAI may improve explanation wording; it cannot change the score.

Presentation checklist

- Open the Dashboard and confirm the history cards are populated.
- Run Bank Account Warning and confirm HIGH RISK.
- Open AI Risk Analysis and point to the highlighted evidence.
- Open Investigation and show all four components as completed.
- Open Recommended Action and read the safety checklist.
- Run Safe Message and confirm LOW RISK.

Privacy reminder

For a classroom demo, use the included fictional examples. Avoid pasting real personal, financial, or account information into the application.

06 / BOUNDARIES

A decision aid, not a guarantee.

ScamShield makes suspicious patterns easier to see, but no keyword-based tool can prove that a message is safe or malicious.

Limitations

- Real scams can avoid obvious keywords.
- Legitimate messages can contain urgency or links.
- A LOW score does not guarantee legitimacy.

Future enhancements

- User-configurable scoring profiles.
- Exportable incident reports.
- Approved sender/domain reputation checks.

Closing message

Pause. Verify. Stay in control.

ScamShield gives users a moment of clarity when a message is designed to create pressure.